Surya Reddy Nallamilli

Chicago, IL | 312-545-7413 | nallamillisuryareddy18@gmail.com | LinkedIn | GitHub | Portfolio

SUMMARY

Software Engineer with 3.5+ years of experience building scalable fintech and healthcare systems using Java, Spring Boot, Python, and Django. Proven expertise in AI-powered automation, batch processing, secure distributed systems, including LLM integration, OCR pipelines, and high-volume transaction workflows.

WORK EXPERIENCE

Bectran, Inc — **June 2025 - Present**
Software Engineer — Schaumburg, IL

- Implemented Gemini LLM-powered fraud detection in the credit-app system using Java, Spring Boot, and AWS to generate automated validation reports with fraud scores, summaries, and follow-up actions.
- Engineered a scheduled invoice mailing system using Lob API with configurable delivery frequencies, daily batch processing, S3-backed PDFs, webhook tracking, and automated dispatch via Spring Boot, handling 40,000+ invoices monthly.
- Developed a financial statement OCR pipeline leveraging Gemini LLM to extract and populate financial data fields automatically, reducing manual entry time by 90%.
- Built a custom W-9 generator within the credit-app workflow using Spring Boot and iText PDF to auto-populate tax details and produce pre-filled forms, reducing manual entry by 70%.

University of Illinois Chicago — **May 2024 - May 2025**
Full Stack Developer — Chicago, IL

- Collaborated on an NIH-funded research initiative to develop a secure, scalable, and HIPAA-compliant Django platform for managing and analyzing confidential patient health data.
- Architected a federated multi-site system connecting 5 medical centers through JWT-based authentication, encrypted key exchange, and secure federated search queries, improving cross-site data accessibility by 70%.
- Implemented a custom Role-Based Access Control (RBAC) framework with user-level permissions and dynamic role customization, enhancing security and administrative flexibility.
- Automated HIPAA-compliant DICOM de-identification for 10,000+ medical images with pixel-level anonymization and metadata cleansing, increasing data processing throughput by 70%.
- Built a dynamic form-builder module allowing researchers to configure fields, validation rules, and privacy flags for sensitive data, streamlining research setup time by 40%.
- Deployed the application on AWS using EC2, ALB, and IAM-based access controls, ensuring SOC 2 compliance.

Cognizant Technology Solutions — **August 2021 - June 2023**
Software Engineer — Hyderabad, India

- Created a reusable SAP framework reducing repetitive ABAP coding by 90%, improving delivery speed and maintainability.
- Designed and implemented large-scale Python data pipelines using Pandas and NumPy, improving data quality by 60%.
- Built unit and integration tests and analyzed SAP production logs to validate business logic, resolve failures, and improve delivery efficiency by 15%.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, JavaScript, C++, Django, Spring Boot, React, Node.js, Express, Flask, REST APIs
Cloud and DevOps: AWS (EC2, S3, Lambda, RDS), Docker, Kubernetes, Nginx, Git, GitHub, Windows IIS, Waitress
Databases & Data Tools: PostgreSQL, MySQL, MongoDB, Oracle, Pandas, NumPy, ELK Stack
Other Skills: Agile, JWT Authentication, LangChain, LLM Integration, Performance Optimization, Datadog Monitoring

EDUCATION

University of Illinois Chicago — **August 2023 - May 2025**
Master of Science in Computer Science | **GPA: 3.89/4.0** — Chicago, IL
Coursework: Machine Learning on Graphs, NLP, Data Science, Deep Learning for CV, Big Data Mining, Distributed Systems.

Amrita Vishwa Vidyapeetham — **July 2017 - June 2021**
Bachelor of Technology in Electronics and Communication Engineering | **GPA: 8.26/10.00** — Coimbatore, India
Coursework: Data Structures and Algorithms, Database Systems, Neural Networks, Software Engineering, Cloud Computing.

PROJECT

AI-Powered Incident and Root Cause Analysis Platform

- Built a Django-based observability platform to ingest service logs, detect log-driven incidents using time-windowed aggregation, and group related failures, reducing manual triage by 60%.
- Integrated OpenAI to generate structured, evidence-backed RCA reports with remediation recommendations and confidence scoring, and built a React dashboard to visualize incidents, achieving 80% accuracy across 25+ simulated incidents.